Kabir Rohit Khanna

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Education

Sep '25 – Present	**PhD in Physics, University of Geneva**, Geneva, Switzerland. Advisor: **Romain Vasseur** [*Relocated*].
Sep '23 – Sep '25	**PhD in Physics, University of Massachusetts Amherst**, Amherst, Massachusetts. Advisor: **Romain Vasseur** .
Oct '22 – Jun '23	**MSc Mathematical and Theoretical Physics, University of Oxford**, Wadham College, Oxford, UK. **Overall grade:** Distinction. **Thesis:** *Long Range Entanglement via Anomalies From Measurements*.
Aug '17 – May '22	**Dual Degree, Indian Institute of Technology Madras**, Chennai, Tamil Nadu. B.Tech. in Engineering Design + M.Tech. in Quantum Science & Technology. **CGPA:** 9.41/10; **Rank:** 3/57. **Thesis:** *Aspects of Quantum Information in Curved Spacetimes*.

Publications and Preprints

2026	**Universal Statistics of Measurement-Induced Entanglement in Tomonaga-Luttinger Liquids**, **Kabir Khanna** and Romain Vasseur, *SciPost Phys.* **21**, 100 (2026). [doi] [arXiv]
2026	**Postselected Criticality in Measurement-Induced Phase Transitions**, Dolly Nambi, **Kabir Khanna**, Andrew Allocca, Thomas Iadecola, Ciarán Hickey, Romain Vasseur, and Justin H. Wilson, *Phys. Rev. Lett.* **137**, 130403 (2026). [doi] [arXiv]
2026	**Measurement-Induced Entanglement in Conformal Field Theory**, **Kabir Khanna** and Romain Vasseur, *Phys. Rev. Lett.* **136**, 160402 (2026). [doi] [arXiv]
2026	**Random Quantum Circuits with Time-Reversal Symmetry**, **Kabir Khanna**, Abhishek Kumar, Romain Vasseur, and Andreas W. W. Ludwig, *Phys. Rev. Research* **8**, 013165 (2026). [doi] [arXiv]
2026	**Replica Transition at Measurement-Altered Ising Criticality**, **Kabir Khanna**, Sara Murciano, and Romain Vasseur, arXiv:2610.13802 (preprint). [arXiv]
2026	**Theory of Measurement-Altered Criticality**, **Kabir Khanna**, Sara Murciano, and Romain Vasseur, arXiv:2608.05289 (preprint). [arXiv]
2023	**Decoherence due to Spacetime Curvature**, Raghvendra Singh, **Kabir Khanna**, and Dawood Kothawala, arXiv:2302.09038 (preprint). [arXiv]

Invited/Conference Talks, Summer Schools, and Lectures Delivered

Jun 2026	[*Delivered lectures*] **QLANDS Summer School on Non-Equilibrium, Circuits, and Complexity** — *Exercise sessions on statistical mechanics of random quantum circuits*.
May 2026	[*Attended school*] **Les Alps 3** — *Generalized symmetries.*
Jan 2026	[*Delivered talk*] **Swiss Quantum Days** — *Measurement-Induced Entanglement in Gapless Quantum Matter*.
Aug 2025	[*Delivered lectures*] **HRI Allahabad** — *Random Quantum Circuits and Their Statistical Mechanics*.
Aug 2025	[*Delivered talk*] **HRI Allahabad** — *Measurement-Induced Entanglement in Conformal Field Theory*.
Mar 2025	[*Delivered talk*] **APS March Meeting** — *Random Quantum Circuits with Time-Reversal Symmetry*.

Teaching and Service

Oct – Jan 2026	**Teaching Assistant for Classical Mechanics 2**, University of Geneva.
Feb – May 2026	**Teaching Assistant for Theory Modules**, University of Geneva.
Feb – May 2024	**Teaching Assistant for Astronomy 105 – Weather**, UMass Amherst.
Sep – Dec 2023	**Teaching Assistant for Physics 118 – Weather and Our Atmosphere**, UMass Amherst.
Jan – May 2022	**Teaching Assistant for PH3520 Quantum Physics**, IIT Madras.
May – Jul 2018	**Mathematics and Physics Tutor**, Lamdon School, Leh, Ladakh.
Refereeing	*Physical Review Letters*, *Physical Review B*, and *PRX Quantum*.

AWARDS AND HONORS

2023	**Graduate Fellowship**, UMass Amherst.
2022	**OCSI Scholarship**, Oxford Cambridge Society of India.
2021	**DAAD WISE Research Scholarship** (awarded; not accepted).
2018	**Best Volunteer**, National Service Scheme (NSS), IIT Madras.
2017	**Silver medal** for a top-4000 rank in JEE Advanced 2017 (of 1.2 million candidates).
2015	**Best Outgoing Student**, DPS Secunderabad.

Last update: Oct 2026